

Research Design and Methods: Lecture 2

Research Methods

(+ lecture 1 ending) Software engineering as technical work and human activity

The lecture begins by completing the previous discussion about whether software engineering should be understood mainly as a natural science or as a form of human activity. The IEEE definition describes software engineering as the application of a systematic, disciplined, and quantifiable approach to software development, operation, and maintenance.

- Software can be viewed as a technical artifact with structural and technical qualities.
- Software development can also be viewed as an industrial and social activity involving processes, collaboration, usability, tools, and development principles.
- Because software development is performed by people, it raises questions about how human activity can be studied, measured, and evaluated.

Measuring a technical or natural object, such as running software, is relatively straightforward. Measuring software development is harder because it involves people whose actions may be shaped by attitudes, social norms, organizational expectations, previous experiences, rewards, beliefs, fears, habits, and many other factors. A key research issue is whether such influences can be measured, and whether natural science methods are sufficient for studying them.

Agile software development as an example

The Manifesto for Agile Software Development, created in 2001, illustrates the human and organizational nature of software engineering. Its main values emphasize people, collaboration, and adaptation rather than technology alone:

- Individuals and interactions over processes and tools.
- Working software over comprehensive documentation.
- Customer collaboration over contract negotiation.
- Responding to change over following a plan.

This does not mean that the items on the right are unimportant. Rather, agile development gives greater priority to the items on the left. The example shows why software engineering research may need to study humans, organizations, motives, values, expectations, and objectives.

Lecture 2: Research methods and methodology

A research method is a structured process for conducting research. Methodology refers more specifically to the systematic and theoretical study of methods in a field, although the two terms are often used interchangeably.

- Methods define how data are collected and recorded.
- They guide what kinds of data are preferred.
- They specify how analysis is performed and how results are validated.
- They may also include epistemological assumptions and rules for interpretation.

Classifying research approaches

Research methods can be classified in several ways, including basic versus applied research, quantitative versus qualitative research, positivist versus interpretivist research, or according to disciplinary traditions. The lecture focuses first on positivism, interpretivism, and the distinction between quantitative and qualitative research.

Positivism

Positivism is closely associated with natural science. It focuses on phenomena that can be observed, measured, or logically verified. Historically, it rejected metaphysical explanations and emphasized experienced phenomena instead. Behaviorism is one example because it studies observable behavior and measurable reinforcement rather than unobservable mental states.

Interpretivism

Interpretivism responds to the difficulty of studying the social world using only natural science methods. Human interactions, cultural meanings, values, and interpretations cannot always be examined as directly observable objects. Researchers are also part of society, so complete detachment is difficult. Interpretivist research therefore emphasizes meaning, context, and understanding people on their own terms.

Positivist and interpretivist assumptions

- **Positivism:** treats reality as objective, tangible, and singular; aims at explanation, prediction, and control; and seeks general laws.
- **Interpretivism:** treats reality as socially constructed and potentially multiple; aims at contextual understanding; and recognizes that meanings depend on time, culture, values, and situation.

Quantitative and qualitative research

Quantitative research describes and measures phenomena using numbers and statistical analysis. It asks questions such as how many, how often, and to what extent.

Typical methods include closed-ended questionnaires, experiments, correlation analysis, and regression analysis.

Qualitative research aims for depth of understanding. It interprets words, perceptions, experiences, and meanings rather than numbers. Data may come from interviews, case studies, focus groups, open-ended questionnaires, observations, videos, photographs, documents, and other rich materials.

Quantitative research can support statistical validity when large samples are available, but it may offer only a limited view of participants' thoughts and experiences.

Qualitative research can provide rich contextual insight, but its findings are not necessarily statistically generalizable. The best choice depends on the research question and the kind of knowledge being sought.

Research approaches in information technology

Järvinen's 2004 classification is used as a pedagogical tool for distinguishing research approaches in information technology and systems. It divides research into studies of reality, mathematical research, and research that evaluates the utility of artifacts.

- **Mathematical approaches:** prove formal theorems and are common in computational research.
- **Conceptual-analytic approaches:** analyze assumptions, theories, models, and frameworks using existing knowledge and logical reasoning.
- **Theory-testing empirical approaches:** use data to examine whether existing theories or hypotheses hold in reality.
- **Theory-creating empirical approaches:** observe practices and develop theoretical explanations from them.
- **Artifact-oriented approaches:** build or evaluate artificial objects such as systems, methods, models, devices, or practices.

Example: test case selection and prioritization

The lecture presents a study on test case selection and prioritization in software testing. Testing is costly, and complete test coverage is usually impossible because software systems are large and complex. Reducing testing is also risky, as failures after delivery increase costs and damage reputation. The study asked how real-world software organizations choose their test case selection approaches, what practices they use, and why they use them.

Study design and data collection

- The study treated testing as a complex human and organizational activity.
- Because no sufficient prior theory existed, the researchers used qualitative research and grounded theory.

- Data were collected through semi-structured interviews in 12 organizations, questionnaires from managers in 31 organizations, and follow-up interviews with testers or programmers in the 12 focus organizations.
- The organizations varied in size, field, and international scope, providing a broad view of Finnish software development at the time.

Grounded theory analysis

- **Open coding:** interview transcripts were examined, and important items were identified as codes and categories.
- **Axial coding:** relationships and causal conditions among categories were analyzed.
- **Selective coding:** one core category, the applied test case selection approach, was chosen and related to the other categories.

Main findings

The researchers identified two main approaches to test case selection:

- **Risk-based selection:** prioritizes failures that would cause the greatest losses.
- **Design-based selection:** prioritizes the main functionalities the software is intended to provide.

Risk-based selection was associated with limited testing resources, flexible development, direct customer influence, frequent exploratory testing, and the aim of minimizing losses if the product fails. Its main limitation was ensuring sufficient test coverage. Design-based selection was associated with more sufficient resources, plan-driven development, indirect customer influence, less exploratory testing, and the aim of verifying intended functionality. Its main limitation was the cost and management effort required.

Conclusion

The study was empirical and theory-creating. It did not test a predefined theory or produce statistical probabilities; instead, it identified and explained patterns in real-world testing practices. The next lecture will continue with qualitative research.